
Old Exam Questions

AI in Life Sciences

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How this file is organized

Questions are collected from the **SS24** exam (25 June 2024). All items are *multiple-select* — the phrasing “which of the following is/are true” means **one or more** options can be correct, and the answer key lists every correct option. Questions are grouped by topic rather than in their original random exam order; the wording was lightly normalised where the export was messy, but the statements and answer choices are kept intact. The full answer key is at the end — the question boxes themselves are intentionally left blank so the file doubles as a practice set.

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Drug Discovery, Molecules & Representations

Question 1: SMILES strings

Which of the following statements about SMILES is/are correct?

- ☐ SMILES stands for *Simplified Molecular Input Line Entry System*.
- ☐ SMILES strings represent molecules in a linear sequence of characters.
- ☐ SMILES cannot be converted into a molecular graph.
- ☐ SMILES is always unique for each molecule, ensuring no two different molecules share the same SMILES.

Question 2: Reading a SMILES string

Given the molecule *pyruvic acid* (2-oxopropanoic acid), whose condensed structure is a methyl group, then a ketone carbonyl, then a carboxylic acid ($\text{CH}_3\text{-C(=O)-C(=O)-OH}$). Which of the following SMILES strings encode this molecule?

- ☐ O=C(C(=O)O)C
- ☐ CC(=O)C(=O)O
- ☐ CCC(=O)O
- ☐ OCC(=O)O

Question 3: Molecular fingerprints

Which of the following statements about molecular fingerprints is/are correct?

- ☐ A unique molecular graph can be constructed from them.
- ☐ They are binary or numerical representations of molecular structures.
- ☐ They are used for similarity searching and machine learning tasks.
- ☐ They provide a 3D representation of molecules.

Question 4: Bioassays

Which of the following statements about bioassays is/are true?

- ☐ They can be used to measure the biological response or function elicited by a substance.
- ☐ They typically test the substance at various concentrations.
- ☐ They are wet-lab procedures to acquire knowledge of a compound in a biological context.
- ☐ They are typically tested *in vitro*, i.e. in a living organism.

Question 5: ADME properties

Which of the following statements about ADME (Absorption, Distribution, Metabolism, and Excretion) properties is/are true?

- ☐ Lipinski's rule of five can be used as a proxy for ADME properties.

- ☐ Good ADME properties guarantee that a compound will be an effective drug.
- ☐ ADME properties such as clearance are often difficult to model.
- ☐ ADME properties are critical in determining the pharmacokinetics of a drug.

Question 6: Characteristics of QSAR / bioactivity data

Which of the following statements regarding the characteristics of QSAR / bioactivity data is/are accurate?

- ☐ QSAR datasets usually have data points in the order of thousands or millions.
- ☐ QSAR datasets typically have a balanced distribution of active and inactive compounds.
- ☐ QSAR datasets often contain series of structurally related compounds.
- ☐ QSAR data often includes noise due to biological variability and experimental inconsistencies.

Question 7: Applications of QSAR modeling

Which of the following applications is/are relevant for QSAR modeling?

- ☐ Creating 3D protein structure models.
- ☐ Identifying potential drug candidates.
- ☐ Predicting the toxicity of new chemical compounds.
- ☐ Mapping the genome sequences of organisms.

Virtual Screening & QSAR Modeling

Question 8: Ligand-based virtual screening

Which of the following statements about ligand-based virtual screening is/are correct?

- ☐ Molecular fingerprints are commonly used in ligand-based virtual screening.
- ☐ Ligand-based virtual screening is not suitable for predicting the bioactivity of unknown compounds.
- ☐ Ligand-based virtual screening requires a 3D structure of the target protein.
- ☐ Ligand-based virtual screening typically only considers bioactivity predictions to select compounds.

Question 9: Data splits in QSAR model evaluation

Which of the following statements about data splits in QSAR model evaluation is/are true?

- ☐ Temporal splits help evaluate the model's performance on future, unseen data by simulating real-world data acquisition over time.
- ☐ Generally, a random split is the best method for evaluating QSAR models as it eliminates biases.

- ☐ Cross-validation is rarely used in the evaluation of QSAR due to computational cost.
- ☐ Cluster splits group similar compounds together, ensuring that structurally similar compounds are not both in the training and test sets.

Question 10: Early-recognition metrics in virtual screening

In virtual screening, the quality of the top-ranked predictions is crucial for efficient drug discovery. Which of the following methods are specifically adapted to prioritise high-quality early hits in the screening process?

- ☐ Precision-Recall Curve.
- ☐ BEDROC (Boltzmann-Enhanced Discrimination of ROC).
- ☐ Average Precision (AP).
- ☐ Enrichment Factor (EF).

Question 11: Target identification / deconvolution

Which of the following statements about target identification / deconvolution is/are correct?

- ☐ For a given molecule, it consists of identifying to which protein it will most likely bind if multiple are present simultaneously.
- ☐ Multi-class classification can be used for it.
- ☐ Multi-task classification can be used for it.
- ☐ For a given molecule, it consists of identifying all the proteins it will bind if multiple are present simultaneously.

Question 12: Drug synergy and antagonism

Which of the following statements about synergy / antagonism is/are true?

- ☐ Drug synergy occurs when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☐ The Loewe additivity model is commonly used to assess drug synergy.
- ☐ Antagonistic effects occur when the combined effect of two or more drugs is greater than the sum of their individual effects.
- ☐ Drug synergy / antagonism evaluation is only applicable to chemically similar drugs.

Question 13: Random Forests vs. Graph Neural Networks

Which of the following statements about the use of Random Forest (RF) and Graph Neural Networks (GNNs) is/are true?

- ☐ GNNs are inherently better for QSAR modeling because they naturally capture the graph-like structure of molecules, leading to more accurate predictions in all cases.
- ☐ RF models are less prone to overfitting compared to GNNs, making them a more reliable choice for datasets with a large number of features but limited samples.
- ☐ RF models are typically more interpretable than GNNs, making it easier to understand

the contribution of individual molecular features to the model's predictions.

- ☐ GNNs are less sensitive to the quality and variability of the input data compared to RFs, making them more consistent in performance across different datasets.

Graph Neural Networks & Advanced Methods

Question 14: Molecular graphs and graph neural networks

Which of the following statements about molecular graphs and graph neural networks is/are true?

- ☐ A molecular graph can be represented by two matrices: one storing the node features and another storing the edges between the nodes.
- ☐ Node features can include atomic number, hybridization state, and formal charge.
- ☐ In a GNN only the node features are used to make the final predictions.
- ☐ The information about the bond type (e.g. single bond, double bond) cannot be included in a GNN.

Question 15: Limitations of GNNs in molecular modeling

Which of the following statements about the limitations of GNNs in the context of molecular modeling is/are true?

- ☐ GNNs are incapable of predicting any molecular properties accurately.
- ☐ GNNs can have difficulty capturing long-range interactions between atoms.
- ☐ GNNs may struggle with very large and complex molecular graphs.
- ☐ GNNs require a significant amount of computational resources for training.

Question 16: Multi-modal contrastive learning

Which of the following statements about multi-modal contrastive learning models is/are true?

- ☐ Multi-modal contrastive learning encodes multiple modalities (e.g. molecules and proteins) into a shared latent space.
- ☐ Multi-modal contrastive learning is a self-supervised learning technique.
- ☐ Multi-modal contrastive learning concentrates (latent) representations from multiple modalities (e.g. molecules and proteins) to make predictions.
- ☐ Multi-modal contrastive learning is an unsupervised learning technique.

Question 17: Multiple instance learning (MIL)

Which of the following statements about multiple instance learning (MIL) is/are true?

- ☐ MIL involves training on sets or bags of instances where each instance is individually labelled.

- ☐ MIL requires the set/bag size to be fixed.
- ☐ MIL involves training on sets or bags of instances that have a single label.
- ☐ MIL models should be invariant to the permutation of instances.

Generative Models & Chemical Synthesis

Question 18: Generative models for small molecules

Which of the following statements about generative models for small molecules is/are true?

- ☐ As the syntax of SMILES strings is complicated and with long dependencies, generators of SMILES representations will produce less than 10% valid SMILES strings.
- ☐ Generative models are used to explore the known chemical space of synthesized molecules.
- ☐ Generative adversarial networks cannot be used to generate small molecules.
- ☐ Autoregressive models can be used to generate SMILES representations of small molecules.

Question 19: Goal-directed molecule generation

Which of the following statements about goal-directed molecule generation is/are true?

- ☐ Goal-directed generation focuses only on replicating existing molecular structures with desired properties.
- ☐ Goal-directed generation aims to create molecules with similar properties to the training set.
- ☐ Reinforcement learning can be used to optimize goal-directed molecule generation.
- ☐ Goal-directed generation aims to create molecules with specific desired properties.

Question 20: Fréchet ChemNet Distance (FCD)

Which of the following statements about Fréchet ChemNet Distance (FCD) and its calculation is/are true?

- ☐ FCD measures the distance between individual atoms of molecules.
- ☐ FCD is calculated by comparing the means and covariances of the latent representations of molecules.
- ☐ FCD is calculated directly on the raw molecular structures without any transformations.
- ☐ FCD is a metric used to compare the distribution of generated molecules to that of real molecules.

Question 21: AI for chemical reactions

Which of the following statements about AI for chemical reactions is/are true?

- ☐ Some methods rely on so-called reaction templates, which are transformation rules for molecules.

- ☐ Deep learning methods that predict outcomes of chemical reactions can be trained on databases containing known chemical reactions.
- ☐ Transformer architectures have been used to predict outcomes of chemical reactions.
- ☐ Predicting chemical reactions with machine learning requires data from quantum mechanical calculations due to bond changes and electron movements.

Question 22: Synthesis planning

Which of the following statements about synthesis planning is/are true?

- ☐ Synthesis planning involves designing a sequence of chemical reactions to produce a target compound.
- ☐ Retrosynthetic analysis is a key technique used in synthesis planning to break down complex molecules into simpler starting materials.
- ☐ Synthesis planning has been made trivial by modern deep learning methods.
- ☐ Synthesis planning involves only selecting the reactants.

Question 23: Retrosynthesis as a reinforcement-learning game

Multi-step retrosynthesis can be formulated as a game that can be played with reinforcement learning. Which of the following statements about the “retrosynthesis game” is/are true?

- ☐ The actions in the retrosynthesis “game” involve selecting reactants for the reactions.
- ☐ The state in the retrosynthesis “game” refers to the set of available building blocks.
- ☐ The environment in the retrosynthesis “game” performs the reaction and returns new molecules.
- ☐ The reward in the retrosynthesis “game” is obtained when the product is successfully synthesized from the building blocks.

Medical Data & Imaging

Question 24: Tabular medical data: types and missing values

Assume we are given a table of patient medical records with three columns: “smoking status” (one of three string values: non-smoker, former smoker, current smoker), “cholesterol level” (floating-point measurements in mg/dL), and “diabetes risk score” (integer values in $[0, 10]$ indicating the likelihood of developing diabetes). Which statements are correct?

- ☐ Missing values in “cholesterol level” can be replaced by the column’s median value to handle skewness.
- ☐ Column “smoking status” is continuous, as it captures the various smoking habits of patients.
- ☐ “cholesterol level” is ordinal since cholesterol measurements can be arranged in increasing order.
- ☐ Missing values in “smoker status” can be replaced by the column’s mean value to handle skewness.

Question 25: Batch effects in medical data

Which of the following statements about batch effects in medical data is/are true?

- ☐ Ignoring batch effects can lead to models that distinguish between batches rather than the actual biological conditions.
- ☐ Batch effects can be eliminated by simply increasing the sample size.
- ☐ Batch effects can create artificial differences in data due to variations in experimental conditions.
- ☐ Batch effects are usually beneficial as they introduce necessary variability into the data.

Question 26: Domain shifts in medical and healthcare data

Data from the medical and healthcare sectors is often affected by domain shifts. Using the lecture notation (y = disease status, e.g. “healthy”/“diseased”; x = patient features, e.g. age, blood parameters, heart rate), which of the following statements about domain shifts are true?

- ☐ If a machine learning model has a high predictive quality on a randomly drawn validation set, this means it will automatically adjust to domain shifts and maintain predictive quality over time.
- ☐ The distribution of the patient features x with distribution $p(x)$ is never affected by domain shifts.
- ☐ During disease waves of infectious diseases (such as COVID-19), the prevalence of a disease and thus the probability of observing this phenotype $p(y)$ changes.
- ☐ Assume that the diagnostic process to determine the disease status y is changed, e.g. an antigen test is replaced by a PCR test. This represents a concept shift $p(y | x)$.

Question 27: Medical imaging modalities

Which of the following statements about medical imaging modalities is/are true?

- ☐ Computed Tomography (CT) scans use ultrasound images to generate images of the body.
- ☐ X-rays are used to differentiate between tissues of varying densities, such as bone and soft tissue.
- ☐ Magnetic Resonance Imaging (MRI) provides detailed images of internal structures by using strong magnetic fields and radio waves.
- ☐ Optical coherence tomography imaging uses multiple X-rays to create images of organs and tissues.

Question 28: Most common task in medical imaging

Which of the following is the most common task in medical imaging compared to typical vision tasks?

- ☐ Image generation.
- ☐ Classification.
- ☐ Segmentation.

- ☐ Object detection.

Question 29: The U-Net architecture

Which of the following statements about the U-Net architecture are correct?

- ☐ The U-Net was the first vision model where the encoder is based on self-attention instead of convolutions.
- ☐ The U-Net architecture uses a symmetrical encoder and decoder.
- ☐ The U-Net has been specifically developed for the object detection task.
- ☐ To aid the decoding process, feature maps from the encoder are concatenated with feature maps of the corresponding up-sampling level of the decoder.

Question 30: The “Clever Hans” effect

What is the “Clever Hans” effect in the context of AI and medical imaging?

- ☐ A bias occurring when enhancing the resolution of low-quality medical images.
- ☐ The situation where an AI model appears to perform well but is actually relying on irrelevant features or artifacts in the data.
- ☐ The phenomenon where an AI system incorrectly predicts outcomes based on biased training data.
- ☐ The phenomenon where data augmentation is used to improve the robustness of AI models in medical imaging.

Question 31: Explainable AI and interpretability in medical imaging

Which of the following statements about explainable AI and interpretability in medical imaging is/are true?

- ☐ Gradient integration utilizes the propagation of a relevance value back to the input layers, layer by layer, to estimate the relevance of each input feature.
- ☐ Explainable AI techniques, such as Grad-CAM, help visualize which parts of a medical image are most important for a model’s prediction.
- ☐ Occlusion analysis involves removing patches of the training data images to improve the model’s interpretability.
- ☐ Counterfactual explanations in medical imaging involve generating alternative images to show what changes would have been needed for a different diagnosis.

Question 32: AlphaFold 2

Which of the following statements about AlphaFold 2 are true?

- ☐ AlphaFold 2 incorporates quantum mechanical calculations to improve the accuracy of its predictions.
- ☐ AlphaFold 2 incorporates multiple sequence alignments and homologous protein structures to improve the accuracy of its predictions.
- ☐ AlphaFold 2 is a deep learning model designed to predict the three-dimensional structure

of proteins from their amino acid sequences.

- ☐ AlphaFold 2 uses a convolutional neural network architecture.

Biological Sequences

Question 33: Composition of biological sequences

Which of the following statements about the composition of biological sequences is/are true?

- ☐ Protein sequences are composed of chains of amino acids linked by peptide bonds.
- ☐ DNA sequences are composed of nucleic acids.
- ☐ Biological sequences do not follow any specific patterns or rules in their composition.
- ☐ RNA sequences are composed of lipids.

Question 34: Deep learning for biological sequences

Which of the following statements about deep learning techniques used for analyzing biological sequences is/are true?

- ☐ Feature extraction from biological sequences can be used to identify motifs, which are recurring patterns or elements within the sequences.
- ☐ One-hot encoding is commonly used to prepare biological sequences as input for deep learning models.
- ☐ Deep learning techniques can be applied to tasks such as classification, segmentation, and generative modeling of biological sequences.
- ☐ 1D-Convolutions, Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and Transformers are all used to analyze biological sequences.

Answer Key

Note on multiple-select scoring

Every question is multiple-select: each listed letter below is a correct option, and any option *not* listed is incorrect. On the original Moodle exam, each correct tick added points and each incorrect tick subtracted them.

- Q1. A, B** — SMILES is the *Simplified Molecular Input Line Entry System* and is a linear character sequence. It *can* be converted to a graph, and is *not* unique (one molecule has many valid SMILES).
- Q2. A, B** — both O=C(C(=O)O)C and CC(=O)C(=O)O encode CH3-CO-COOH. CCC000 and OCC00 do not.
- Q3. B, C** — fingerprints are binary/numerical descriptors used for similarity search and ML. They are lossy (no unique graph reconstruction) and carry no 3D information.
- Q4. A, B, C** — bioassays measure a biological response, test at various concentrations (dose-response), and are wet-lab procedures. “*in vitro* = in a living organism” is false (*in vivo* is the living organism).
- Q5. A, C, D** — Lipinski’s Ro5 is an ADME proxy, clearance is hard to model, and ADME is central to pharmacokinetics. Good ADME does *not* guarantee an effective drug.
- Q6. C, D** — QSAR data often comes in structurally related series and is noisy. It is usually *small* (not thousands/millions) and *imbalanced* (not balanced).
- Q7. B, C** — QSAR predicts drug candidates and toxicity. 3D protein modeling and genome mapping are different problems.
- Q8. A** — fingerprints are standard in ligand-based VS. It *is* suitable for unknown compounds, does *not* need the protein structure, and considers more than bioactivity (ADME, filters).
- Q9. A, D** — temporal splits simulate real-world acquisition; cluster splits keep structurally similar compounds out of both sets. Random splits do *not* eliminate bias, and CV *is* used.
- Q10. B, D** — BEDROC and the Enrichment Factor reward early recognition. Precision-Recall curve and Average Precision are general metrics, not early-recognition-specific.
- Q11. A, B** — target identification finds the *most likely* target via multi-class classification. “All proteins” and “multi-task” describe target *prediction*, not identification.
- Q12. A, B** — synergy is “greater than the sum”; the Loewe additivity model assesses it. Antagonism is *less* than the sum, and synergy applies to any drug pair (not only similar ones).
- Q13. B, C** — RF overfits less and is more interpretable than GNNs. GNNs are *not* universally better and are *not* less sensitive to data quality.
- Q14. A, B** — a molecular graph is a node-feature matrix plus an edge/adjacency matrix; node features include atomic number, hybridization, charge. GNNs *do* use edge information, and bond type *can* be encoded.
- Q15. B, C, D** — GNNs struggle with long-range interactions, very large graphs, and are compute-heavy. They are *not* “incapable of predicting any property accurately”.

- Q16. A, B** — multi-modal contrastive learning maps modalities into a shared latent space and is *self-supervised*. It does not “concentrate representations to make predictions”, and is *not* unsupervised.
- Q17. C, D** — a MIL bag has a *single* label and the model must be permutation-invariant. Instances are *not* individually labelled, and bag size need *not* be fixed.
- Q18. D** — autoregressive models can generate SMILES. Modern generators produce mostly valid SMILES, generative models explore *novel* space, and GANs *can* generate molecules.
- Q19. C, D** — goal-directed generation targets *specific* desired properties and can be optimised with RL. Replicating existing structures / matching the training distribution describes *distribution learning*.
- Q20. B, D** — FCD compares means and covariances of *latent* (ChemNet) representations and measures distance between *distributions* of generated vs. real molecules — not individual atoms, not raw structures.
- Q21. A, B, C** — reaction templates, training on known-reaction databases, and transformer architectures are all used. QM data is *not* required.
- Q22. A, B** — synthesis planning designs a reaction sequence and uses retrosynthetic analysis. It is *not* trivial, and involves more than just picking reactants.
- Q23. A, C, D** — per the lecture’s framing the *action* is selecting reactants, the *environment* applies the reaction and returns new molecules, and the *reward* comes when the current set are all available building blocks. Only the *state* option is wrong: the state is the *current set of molecules* to be made, not the set of available building blocks.
- Q24. A** — cholesterol is numeric, so the *median* handles skewed missing values. Smoking status is categorical (not continuous/mean-imputable), and cholesterol is continuous, not merely ordinal.
- Q25. A, C** — ignored batch effects make models separate batches instead of biology, and they create artificial differences. They are *not* removed by more samples and are *not* beneficial.
- Q26. C, D** — changing prevalence is a label/prior shift $p(y)$; changing the diagnostic procedure is a concept shift $p(y | x)$. Good validation accuracy does *not* auto-adapt, and $p(x)$ *can* shift.
- Q27. B, C** — X-rays distinguish tissue densities; MRI uses magnetic fields and radio waves. CT uses X-rays (not ultrasound), and OCT uses light (not X-rays).
- Q28. B, C** — classification and segmentation are the dominant medical-imaging tasks (segmentation especially, vs. typical vision tasks).
- Q29. B, D** — U-Net has a symmetric encoder/decoder with skip connections (encoder feature maps concatenated into the decoder). It is convolutional (not attention-based) and was built for *segmentation*, not detection.
- Q30. B** — the Clever Hans effect is a model that *looks* good but relies on irrelevant features/artifacts (shortcut learning).
- Q31. B, D** — Grad-CAM highlights important image regions; counterfactuals show what change would flip the diagnosis. The first option mislabels LRP as “gradient integration”, and occlusion perturbs *input*, not training data.
- Q32. B, C** — AlphaFold 2 uses multiple sequence alignments + homologous structures to predict 3D structure from sequence. It uses an attention-based (Evoformer) architecture, *not* a CNN, and no QM calculations.

- Q33. A, B** — proteins = amino acids via peptide bonds; DNA = nucleic acids. Sequences *do* follow rules, and RNA is nucleic acid, not lipid.
- Q34. A, B, C, D** — all true: motif/feature extraction, one-hot input encoding, classification/segmentation/generation tasks, and 1D-CNN / RNN / LSTM / Transformer architectures.